

Chapter 4

1. Generators $\mathbb{Z}_6 = U(6) = \{1, 5\}$

Generators $\mathbb{Z}_8 = U(8) = \{1, 3, 5, 7\}$

Generators $\mathbb{Z}_{20} = U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}$

2. $|\langle a \rangle|, |\langle b \rangle|, |\langle c \rangle| = 6, 8, 20$

$\langle a \rangle = \langle a^j \rangle$ when $\gcd(6, j) = 1$

This means generators of $\langle a \rangle = \{a^i \mid \gcd(6, i) = 1\}$ Possible values for $j \leq 6$ that satisfies this = $\{1, 5\}$ Thus generators are a^1 and a^5

ii. $\langle b \rangle = \langle b^j \rangle$ when $\gcd(8, j) = 1$

This means generators of $\langle b \rangle = \{b^i \mid \gcd(8, i) = 1\}$

iii. Possible values for $j \leq 8$ that satisfies this = $\{1, 3, 5, 7\}$

Thus generators are b^1, b^3, b^5, b^7

iii. $\langle c \rangle = \langle c^j \rangle$ when $\gcd(20, j) = 1$

This means generators of $\langle c \rangle = \{c^i \mid \gcd(20, i) = 1\}$ Possible values for $j \leq 20$ that satisfies this = $\{1, 3, 7, 9, 11, 13, 17, 19\}$ Thus generators are $c^1, c^3, c^7, c^9, c^{11}, c^{13}, c^{17}, c^{19}$

3. $\langle 20 \rangle = \{20, 10, 0\}$

$\langle 10 \rangle = \{10, 20, 0\}$

ii. Let $|a| = 30$. Find elements of subgroups $\langle a^{20} \rangle$ and $\langle a^{10} \rangle$

$|\langle a^{20} \rangle| = 30 / \gcd(30, 20) = 30/10 = 3$

Since $|a^k| = |\langle a^k \rangle|$ for any $k \in \mathbb{Z}$, $\langle a^{20} \rangle$ has order 3

$\langle a^{20} \rangle = \{a^{20}, a^{40}, a^{60}\}$

Since $a^{30} = e$, given $|a| = 30$, we reduce to modulo 30

$a^{10} = a^{10}, a^{60} = a^0 = e$

$\therefore \langle a^{20} \rangle = \{a^{20}, a^{40}, e\}$

$$|a^3| = 30 / \gcd(30, 10) = 30/10 = 3$$

Since $|a^k| = |\langle a^k \rangle|$ for any $k \in \mathbb{Z}$, $\langle a^3 \rangle$ has order 3

$$\langle a^3 \rangle = \{a^3, a^6, a^9\}$$

Since $a^{30} = e$ given $|a| = 30$, we reduce to modulo 30

$$\langle a^3 \rangle = \{a^3, a^6, e\}$$

5. $\langle 3 \rangle = \{3, 4, 7, 1\}$

$$\langle 7 \rangle = \{7, 9, 3, 4\}$$

4. $\langle 3 \rangle = \{3, 6, 9, 12, 15, 0\}$

$$\langle 15 \rangle = \{15, 12, 9, 6, 3, 0\}$$

Let $|a| = 18$

$$|a^3| = 18 / \gcd(18, 3) = 18/3 = 6$$

Since $|a^k| = |\langle a^k \rangle|$ for any $k \in \mathbb{Z}$, $\langle a^3 \rangle$ has order 6

$$\langle a^3 \rangle = \{a^3, a^6, a^9, a^{12}, a^{15}, a^{18}\}$$

Since $a^{18} = e$ given $|a| = 18$, we reduce to modulo 18

$$\therefore \langle a^3 \rangle = \{a^3, a^6, a^9, a^{12}, a^{15}, e\}$$

$$|a^6| = 18 / \gcd(18, 6) = 18/3 = 6$$

Since $|a^k| = |\langle a^k \rangle|$ for any $k \in \mathbb{Z}$, $\langle a^6 \rangle$ has order 6, just like $\langle a^3 \rangle$

$$\langle a^6 \rangle = \{a^6, a^{12}, a^{18}, a^{24}, a^{30}, a^{36}\}$$

Since $a^{18} = e$, we can reduce to modulo 18

$$\therefore \langle a^6 \rangle = \{a^6, a^{12}, a^{18}, a^{24}, a^{30}, e\}$$

$$\therefore \langle a^3 \rangle = \langle a^6 \rangle \text{ in this case}$$

6. If $|a| = n$. Then $\langle a^i \rangle = \langle a^j \rangle$ iff $\gcd(n, i) = \gcd(n, j)$

7. Let $|a| = 15$

a. $|a^3| = 15 / \gcd(15, 3) = 15/3 = 5$

$$|a^6| = 15 / \gcd(15, 6) = 15/3 = 5$$

$$|a^9| = 15 / \gcd(15, 9) = 15/3 = 5$$

$$|a^2| = 15 / \gcd(15, 12) = 15 / 3 = 5$$

$$b. |a^3| = 15 / \gcd(15, 5) = 15 / 5 = 3$$

$$|a^{10}| = 15 / \gcd(15, 10) = 15 / 5 = 3$$

$$c. |a^2| = 15 / \gcd(15, 2) = 15 / 1 = 15$$

$$|a^4| = 15 / \gcd(15, 4) = 15 / 1 = 15$$

$$|a^8| = 15 / \gcd(15, 8) = 15 / 1 = 15$$

$$|a^{14}| = 15 / \gcd(15, 14) = 15 / 1 = 15$$

10 In $\mathbb{Z}_{24}$, observe that $24/8 = 3$ is one option subgroup of order 8. To find others, we know they're of the form $3K$, where $\gcd(8, K) = 1$. Thus,

$$\langle 3 \rangle = \langle 3 \cdot 3 \rangle = \langle 3 \cdot 5 \rangle = \langle 3 \cdot 7 \rangle$$

$\therefore$ The generators of the subgroup of order 8 in $\mathbb{Z}_{24}$ are $\{3, 9, 15, 21\}$

b. Let $G = \langle a \rangle$ and let $|a| = 24$ [Let generator for order 24]

We want $|a^k| = 8$ so that $|\langle a^k \rangle| = 8$ also

$$\text{This means } \frac{24}{\gcd(24, k)} = 8 ; \quad \gcd(24, k) = 3$$

Values of k in $\mathbb{Z}_{24}$ that the equation are $k = \{3, 9, 15, 21\}$

$\therefore$ Generator of subgroup of order 8 are: a^3, a^9, a^{15}, a^{21}

11 Let G be any group and $a \in G$ [show $\langle a^{-1} \rangle = \langle a \rangle$]

By definition, we know $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$

Similarly, we know $\langle a^{-1} \rangle = \{(a^{-1})^n \mid n \in \mathbb{Z}\}$

We need to show $[\langle a^{-1} \rangle \subseteq \langle a \rangle \text{ and } \langle a \rangle \subseteq \langle a^{-1} \rangle]$

Any element in $\langle a^{-1} \rangle$ has form $(a^{-1})^n$ for some $n \in \mathbb{Z}$

$$(a^{-1})^n = a^{-n}$$

Now we know $a^{-n} \in \langle a \rangle$ because $\langle a \rangle$ contains all integer powers

of a , $(a^{-1})^n = a^{-n} \in \langle a \rangle$
 Thus, $\langle a^{-1} \rangle \subseteq \langle a \rangle$

Now, consider any element in $\langle a \rangle$. This has form a^n for some $n \in \mathbb{Z}$

We also know $a^n = (a^{-1})^{-n}$

Since $(a^{-1})^{-n} \in \langle a^{-1} \rangle$ because $\langle a^{-1} \rangle$ contains all integer powers of a^{-1} , we can say

$$a^n = (a^{-1})^{-n} \in \langle a^{-1} \rangle$$

Thus, $\langle a \rangle \subseteq \langle a^{-1} \rangle$

∴ Since $\langle a^{-1} \rangle \subseteq \langle a \rangle$ and $\langle a \rangle \subseteq \langle a^{-1} \rangle$
 $\langle a^{-1} \rangle = \langle a \rangle$

12 In $\mathbb{Z}$, we know $\langle 3 \rangle = \{3k \mid k \in \mathbb{Z}\}$

So, $\langle 3 \rangle = \{-12, -9, -6, -3, 0, 3, 6, \dots\}$

From 11 above, we proved $\langle a^{-1} \rangle = \langle a \rangle$

∴ Element -3 can also generate $\langle 3 \rangle$ since it has inverses

Thus, the subgroup $\langle 3 \rangle$ has 2 generators, $3, -3$

b. Let a have infinite order

We know $\langle a^3 \rangle = \{(a^3)^n \mid n \in \mathbb{Z}\}$

$$\langle a^3 \rangle = \{a^0, a^3, a^6, a^9, \dots\}$$

Just like in previous problems, only other element capable of generating the same entire subgroup is $(a^3)^{-1}$

∴ Generators of subgroup $\langle a^3 \rangle$ is a^3 and a^{-3}

13 In $\mathbb{Z}_{24}$, $\langle 21 \rangle \cap \langle 10 \rangle$

$$= \{21, 18, 15, 12, 9, 6, 3, 0\} \cap \{10, 20, 6, 16, 2, 12, 22, 8, 18, 4, 14, 0\}$$

$$= \{0, 6, 12, 18\}$$

$\therefore$ Generator for $\langle 21 \rangle \cap \langle 10 \rangle$ is smallest integer in the set
 has the identity. Generator is 6

* 14 Let $|a| = 24$

We know $\langle a^{21} \rangle = \{a^{21k} \mid k \in \mathbb{Z}\}$

$|a^{21}| = 24 / \gcd(24, 21) = 24/3 = 8$

Thus $\langle a^{21} \rangle$ has 8 elements

$\langle a^{21} \rangle = \{a^{21}, a^{42}, a^{63}, a^{84}, a^{105}, a^{126}, a^{147}, e\}$

We also know $\langle a^{10} \rangle = \{a^{10m} \mid m \in \mathbb{Z}\}$

$|a^{10}| = 24 / \gcd(24, 10) = 24/2 = 12$

Thus $\langle a^{10} \rangle$ has 12 elements

$\langle a^{10} \rangle = \{a^{10}, a^{20}, a^{30}, a^{40}, a^{50}, a^{60}, a^{70}, a^{80}, a^{90}, a^{100}, a^{110}, e\}$

To find intersection $\langle a^{21} \rangle \cap \langle a^{10} \rangle$, we look for element
 that can be expressed in both forms $a^{21k} = a^{10m}$

$a^{21k} = a^{10m}$

$21k = 10m \pmod{24}$

In this case, $\gcd(21, 10) = 1$

so $\langle a^{21} \rangle \cap \langle a^{10} \rangle = \{e\}$

$\therefore$ In general, $\langle a^r \rangle \cap \langle a^s \rangle$ is given by $a^{\gcd(r, s)}$

* 15. Let G be abelian and $H = \{g \in G \mid |g| \text{ divides } 12\}$.
non-empty: H is non-empty because e has order 1 which
 divides 12.

inverses: Let $g \in H$. This means $|g|$ divides 12

Since G is abelian, $|g^{-1}| = |g|$

Since $|g|$ divides 12, $|g^{-1}|$ must also divide 12

closure Let $x, y \in H$ and $|x| = d_1$; $|y| = d_2$

We can express as $d_1 = \frac{12}{m_1}$, $d_2 = \frac{12}{m_2}$

For any group, the order of the product xy divides

least common multiple of orders x & y
 $|xy| \mid \text{L.C.M.}(|x|, |y|)$

Since d_1 and d_2 divide 12, $\text{lcm}(d_1, d_2)$ must also divide 12

$\therefore |xy|$ divides 12

$\therefore H$ is a subgroup of G by the two-step subgroup test

b. The proof can be generalized to any positive integer n . Group structure does not depend on specific n -value, as long as only orders of elements that divide n is being considered

16. Consider all cases

Let $|a| = n$. This means $a^n = e$ and $n \geq 1$

When $|a| = 1$, then $a = e$.

$$a^2 = e^2 = e$$

$$|a^2| = 1$$

$$\therefore |a| = |a^2|$$

When $|a| = 2$, then $a^2 = e$.

$$|a^2| = 1 \text{ because } (a^2)^n = e \text{ when } n = 1$$

$$\therefore |a| \neq |a^2|$$

For any order $n > 2$, $|a| = n$ means a^d has order $\frac{n}{d}$ where $d = \gcd(n, 2)$

Thus $|a| = |a^d|$ where $n = \frac{n}{d}$ and d equal 1

For $d = 1$, $n \neq 2$ must be coprime and this is the criteria for success

$|a| = |a^2|$ if and only if $|a|$ is relatively prime to the power of 2 in this case. If $|a|$ is odd

17. Consider all cases

Let $|a| = n$

We know $|a^n| = \frac{n}{\gcd(n, n)}$

$$\text{Thus } |a^2| = \frac{n}{\gcd(n, 2)}$$

$$|a^{12}| = \frac{n}{\gcd(n, 12)}$$

$$\text{For } |a^2| = |a^{12}|, \quad \frac{n}{\gcd(n, 2)} = \frac{n}{\gcd(n, 12)}$$

Thus $\gcd(n, 2) = \gcd(n, 12)$ iff n is odd, which implies both $\gcd(n, 2)$ & $\gcd(n, 12)$ equal 1

$\therefore |a^2| = |a^{12}|$ iff $|a|$ is odd

19 In a cyclic group generated by a of infinite order, every element can be expressed as a^k for $k \in \mathbb{Z}$

For a^k to have finite order, there must exist a positive integer n , such that $(a^k)^n = e$

$$a^{kn} = e$$

Since a has infinite order, only solution is when $kn = 0$. Thus k must equal 0. Since n is a positive integer

$$\text{So, we have } a^0 = a^n = e$$

$\therefore$ Only element of finite order in a cyclic G with existing elements of infinite order is the identity, e

21 a. $|a|$ must divide 12

b. $|a|$ must divide m

c. Let $|G| = 24$ and $a^8 \neq e$, $a^{12} \neq e$

Since G is cyclic, there exists a generator a , such that $|a| = |G| = 24$

By FTG, we know the order of a must divide the order of the group. So possible orders: 1, 2, 3, 4, 6, 8, 12, 24. We know that 6 and 12 are ruled out

Considering all options

$|a| = 1 \Rightarrow$ contradicts non-identity conditions

$|a| = 2 \Rightarrow$ contradicts $a^{12} \neq e$ since $a^{12} = (a^2)^6 = e$

$|a| = 3 \Rightarrow$ in the same vein contradicts $a^{12} = e$

$|a| = 6 \Rightarrow$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $a^{12} = e$

Only possible order left is 24

So $|a|$ must equal 24

$\therefore$ Since a has order 24, cyclic group $\langle a \rangle$ contains G exactly 24 elements because $|a| = |\langle a \rangle|$

We have G defined as having 24 elements, we can conclude $\langle a \rangle = G$

22. Let G be a group and $|G| = 3$

Using FTCT, the order of any subgroup of G must divide order of G .

Since $|G| = 3$, only possible divisors are 1 and 3.

We can write as $G = \{e, a, b\}$

We know a group is cyclic if there exists an element $g \in G$ such that every element in G can be written in the form g^n for $n \in \mathbb{Z}$

Consider element a . If $|a| = 1$, then a must be the identity element, e . However this means $G = \{e\}$ which contradicts $|G| = 3$

If $|a| = 3$, then $\langle a \rangle = \{e, a, a^2\}$

This implies $\langle a \rangle$ must contain all three elements of G

In this case e, a, a^2 ; in which $a^2 = b$ in G

$\therefore$ Since we showed $\langle a \rangle \subseteq G$, when $|a| = 3$, then $G = \langle a \rangle$ because $|G| = |\langle a \rangle|$

29 Order of element k in $\mathbb{Z}_n$ is defined as the smallest positive integer m , such that $km \equiv 0 \pmod{n}$.
For k to have order 8, $8k \equiv 0 \pmod{8000000}$

One example in $\mathbb{Z}_{8000000}$ is $8000000 | 8 = 1000000$

Now, the other elements with order 8 have the form $1000000L$, where $\gcd(8, L) = 1$

Integers L coprime to 8: $\{1, 3, 5, 7\}$

Elements of order 8 in $\mathbb{Z}_{8000000} = \{1000000, 3000000, 5000000, 7000000\}$

b. Using FTG, we showed there are all unique elements that satisfy $\gcd(L, 8000000) = 1000000$ confirmed to have order 8

31 Let G be a group with $e \in G$

Since G is finite, the order of every element in G must be finite also.

Using FTG, $|a| \mid |G|$

Let S equal set of all orders in G . $S = \{|a| \mid a \in G\}$

Since order of each element divides $|G|$, no element $a \in G$ can possess $|a| > |G|$

Thus S contains positive integer divisors of $|G|$

Now, let $n = \text{L.C.M.}(|a_1|, |a_2|, \dots, |a_n|)$ where $|a_1| = k$

Since n is the least common multiple of all these orders of elements in G , n is divisible by each element's order for all $a_i \in G$

We know there exists integer k , $n = k \cdot |a|$ by L.C.M.
 $a^n = a^{k \cdot |a|} = (a^{|a|})^k$

$$a^n = e^k = e$$

$\therefore a^n = e$ for all $a \in G$

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Let $H = \{n \in \mathbb{Z} \mid n \text{ is divisible by } 8, 10\}$

nonempty: Set H contain 0 because it is divisible by any integer

closure Let $a, b \in H$. This means a is divisible by both 8 and 10; b is divisible by both 8 and 10. This also means these elements are divisible by $\text{LCM}(8, 10) = 40$

All elements in H must be divisible by 40

Thus if a & b are both divisible by 40, their sum $a+b=c$ is also divisible by 40 since c is a multiple of 40. $\therefore a+b \in H$

inverse Let $a \in H$. This means a is divisible by 40

In group $\mathbb{Z}$ here, additive inverse of a is $-a$ which is divisible by 40 because $-a$ is a multiple

of 40

$\therefore -a \in H$

By 2SST, $H = \{n \in \mathbb{Z} \mid n \text{ is divisible by } 8, 10\}$ is a valid subgroup of $\mathbb{Z}$

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